

§11.3 The Integral Test and Estimates of Sums

Homework : 1,4,5,7,12,15,21,25,29,33,38,39

- **Who :** That is what kind of the series can use the integral test to check its convergence?

$a_i :$ (i) essentially positive (ii) decreasing (iii) $f(x) = a_x$ is continuous	(11.3-1)
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- **What :** $\sum_{i=1}^{\infty} a_i$ and $\int_1^{\infty} a_x dx$ both converges or diverges.

Example 1 : $\sum_{n=1}^{\infty} \frac{1}{n^p}$ (p -series).

Recall : $\int_1^{\infty} \frac{1}{x^p} dx = \begin{cases} \text{convergence, } p > 1, \\ \text{divergence, } p \leq 1. \end{cases}$

$\Rightarrow$ **Theorem1 :** $\sum_{n=1}^{\infty} \frac{1}{n^p} = \begin{cases} \text{convergence, } p > 1, \\ \text{divergence, } p \leq 1. \end{cases}$

Example 2 : $\sum_{n=1}^{\infty} \frac{1}{n^2 + 1}$

Solution :

$$\begin{aligned}
 \int_1^{\infty} \frac{1}{x^2 + 1} dx &= \tan^{-1} x \Big|_1^{\infty} \\
 &= \frac{\pi}{2} - \frac{\pi}{4} \\
 &= \frac{\pi}{4} \text{ (convergence)}
 \end{aligned}$$

$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^2 + 1}$ convergence.

Example 3 : $\sum_{n=1}^{\infty} \frac{\ln n}{n}$

Solution :

$$\begin{aligned} \int_1^{\infty} \frac{\ln x}{x} dx &= \int_0^{\infty} u du, \quad (\ln x = u). \\ &\Rightarrow \text{divergence} \\ &\Rightarrow \sum_{n=1}^{\infty} \frac{\ln n}{n} \quad \text{divergence.} \end{aligned}$$

Example 4 : $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^p}$

Solution :

$$\begin{aligned} \int_2^{\infty} \frac{1}{x(\ln x)^p} dx &= \int_{\ln 2}^{\infty} \frac{du}{u^p} \quad (u = \ln x) \\ &\Rightarrow \begin{cases} p \leq 1, \text{ divergence} \\ p > 1, \text{ convergence.} \end{cases} \end{aligned}$$

Example 5 : Discuss the convergence of $\sum_{n=1}^{\infty} b^{\ln n}$.

Solution :

$$\begin{aligned} \sum_{n=1}^{\infty} b^{\ln n} &= \sum_{n=1}^{\infty} (e^{\ln b})^{\ln n} = \sum_{n=1}^{\infty} (e^{\ln n})^{\ln b} = \sum_{n=1}^{\infty} n^{\ln b}. \\ &\Rightarrow \begin{cases} \ln b \geq -1 \Rightarrow b \geq e^{-1} \Rightarrow \text{divergence.} \\ \ln b < -1 \Rightarrow b < e^{-1} \Rightarrow \text{convergence.} \end{cases} \end{aligned}$$

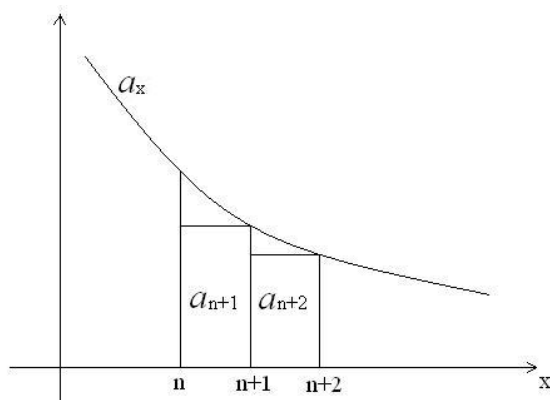
Theorem2 : Let $\{a_i\}$ satisfy 11.3-1, and let $R_n = S - S_n$.

$$\text{Here } S = \sum_{i=1}^{\infty} a_i \quad \text{and} \quad S_n = \sum_{i=1}^n a_i.$$

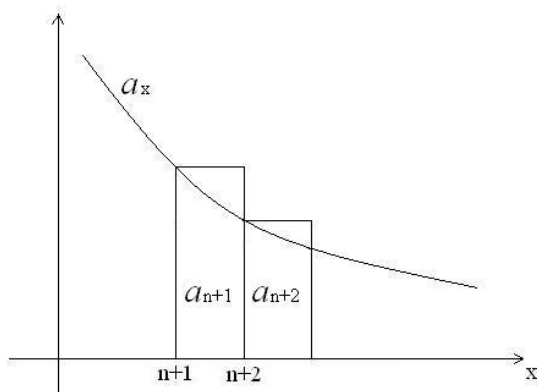
Then.

$$\int_{n+1}^{\infty} a_x dx \leq R_n \leq \int_n^{\infty} a_x dx$$

Proof :



(由上圖得 $R_n = a_{n+1} + a_{n+2} + \dots \leq \int_n^\infty a_x dx$) .



(由上圖得 $R_n \geq \int_{n+1}^\infty a_x dx$) .

Example 6 : Use Theorem 2. to estimate the sum of the series $\sum_{n=1}^\infty \frac{1}{n^3}$ with $n = 10$.

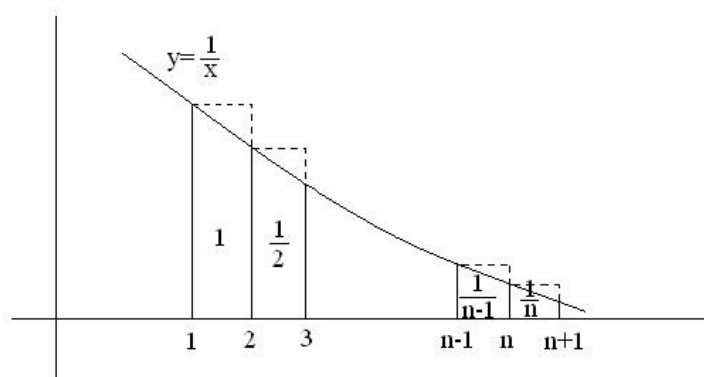
Solution :

$$\begin{aligned} \int_{11}^\infty \frac{1}{x^3} dx &\leq S - S_{10} \leq \int_{10}^\infty \frac{1}{x^3} dx \\ \Rightarrow \frac{1}{2(11)^2} &\leq S - S_{10} \leq \frac{1}{2(10)^2} \\ \Rightarrow S_{10} + \frac{1}{2(11)^2} &\leq S \leq S_{10} + \frac{1}{2(10)^2} \\ S_{10} &\approx 1.197532 \\ \Rightarrow 1.201664 &\leq S \leq 1.202532 \end{aligned}$$

Example 7 : Let $t_n = 1 + \frac{1}{2} + \dots + \frac{1}{n} - \ln n$.

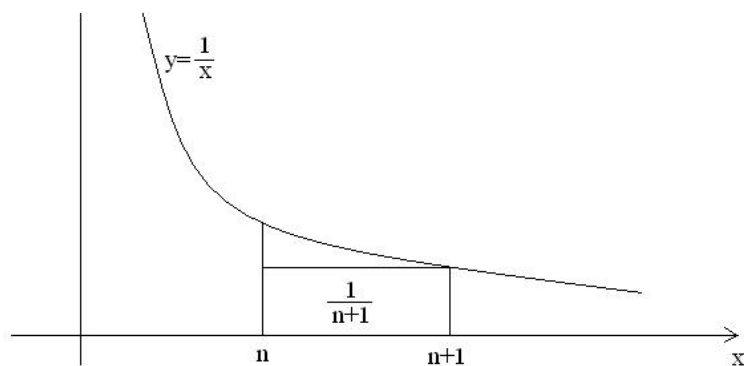
- (a) Show that $t_n > 0$
- (b) Show that $t_n - t_{n+1} > 0$
- (c) Prove that $t_n \leq 1$ for all n .

Proof : (a)



由上圖得知 $t_n > 0$

(b)



由上圖得知 $\ln(n+1) - \ln(n) > \frac{1}{n+1} > 0$

$\Rightarrow t_n - t_{n+1} > 0$

$\Rightarrow t_n$ is a decreasing sequence.

- (c) 由(b)得 , $t_1 \geq t_n$ for all n . 但 $t_1 = 1$
 $\Rightarrow 1 \geq t_n$ for all n .